Supplementary material for Observe, Imagine, Abstain: Hallucination-Aware 3D Reconstruction with Discrete Diffusion and Continual Adaptation

Anonymous CVPR submission

Figure and table numbers below are local to this document; the main paper is self-contained and refers to this material by section title.

Reproducibility. Every number in the paper maps to a CSV and a configuration file in the released log; the benchmark is 700 lines of Python with no training, and all its model runs fit on one 16 GB GPU in about eight hours. The support head trains in about 25 minutes on one GPU from the released configuration; its runs are logged under the P3 run ids.

Disclosure of AI assistance. An AI assistant (Claude, Anthropic) was used under the authors' direction for writing code, orchestrating experiment runs and drafting text. The authors designed the study, chose and froze every hyperparameter, ran and checked every experiment, verified every number against its logged run, and are responsible for all content. The assistant is not an author.

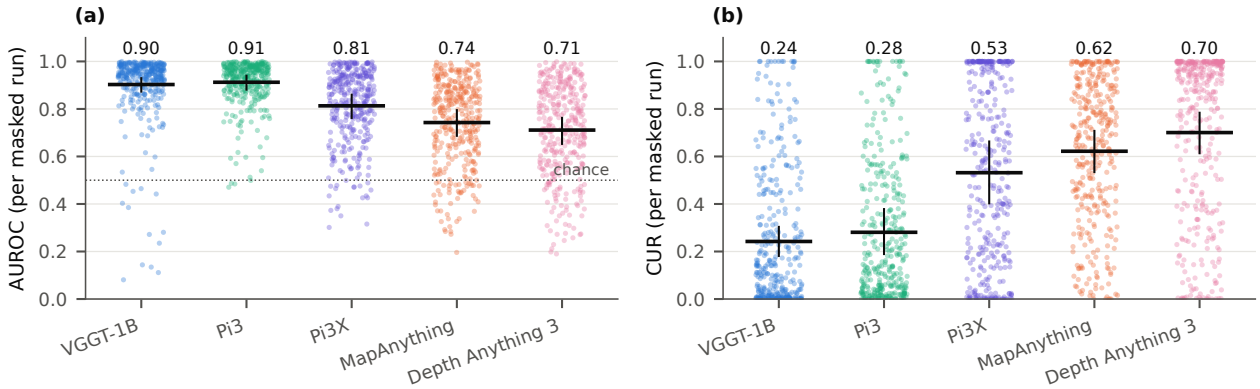


Figure 1. **Per-run spread on DTU.** AUROC and CUR for every (scan, seed, radius) run under blur and noise fills, with pooled means and intervals. The spread across scans is wide for every model; the intervals in the main tables come from resampling scans.

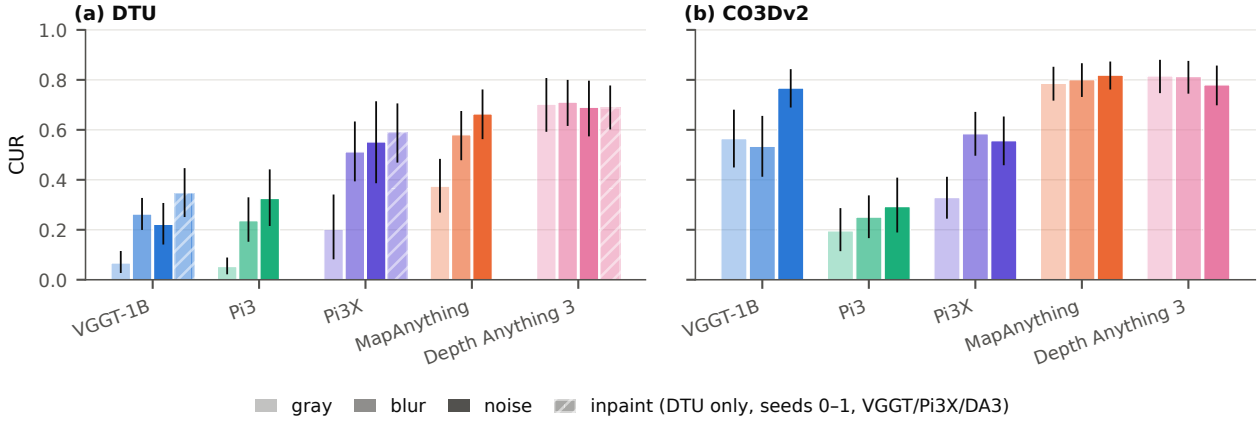


Figure 2. **Fill ablation.** CUR per model and fill on DTU and CO3Dv2 (five seeds) and for the inpainted fill on DTU (two seeds, three models). A visible gray patch gives the lowest leakage; natural-statistics and inpainted fills give more.

Run registry. Every table cell maps to a run id (P1-full, P2-full, P2-stress, P2-co3d, P2-co3d-b, P3-pv-smoke, P2-smoke-b) whose CSV, configuration JSON and kernel log are released with the code, together with the dated log of every failed run and every correction.

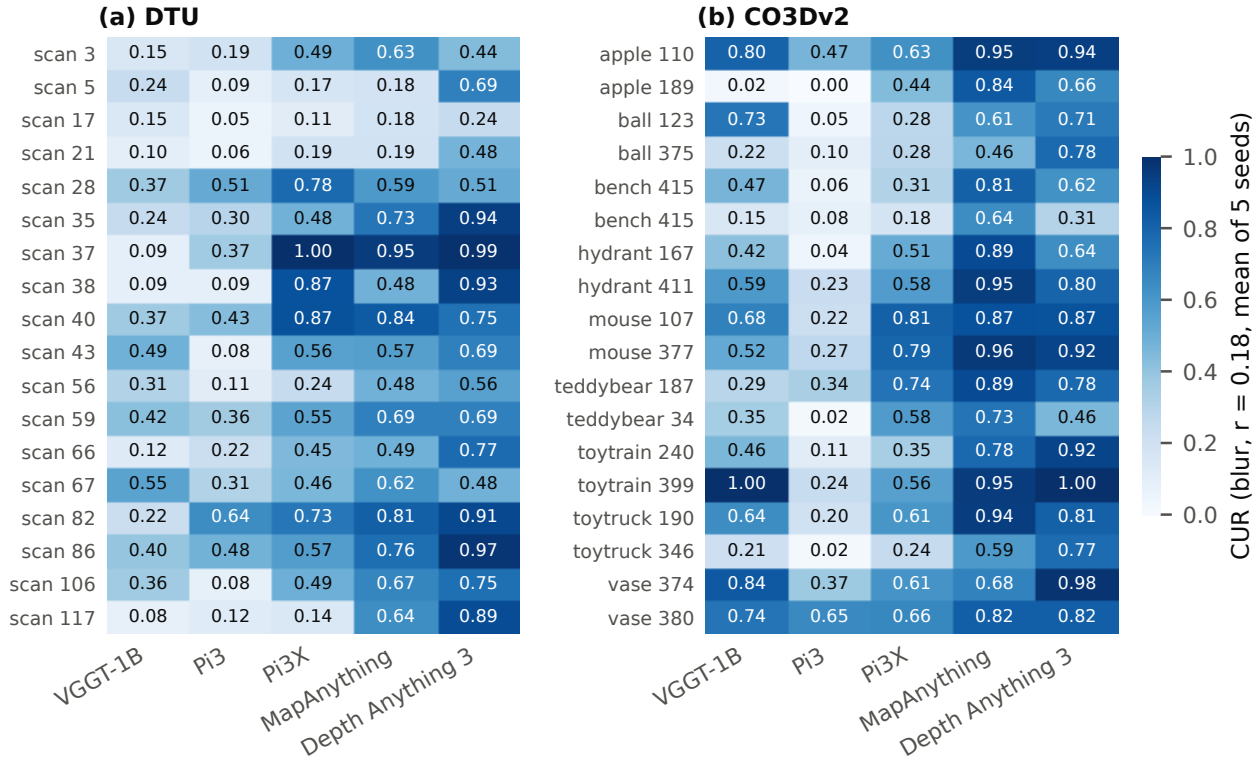


Figure 3. **Per-scene CUR** (blur fill, radius 0.18, mean over seeds) for every model on DTU scans and CO3Dv2 scenes. Rows are scenes, columns are models.